



Spatial Evolution and Collaborative Innovation of China's Lithium-Ion Battery Research and Development Enterprises: Evidence from a National Innovation Network

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Abstract: The spatial configuration and collaborative networks of research and development (R&D) enterprises are continuously reshaping the innovation landscape of strategic industries. Clarifying this evolution is crucial for advancing China's lithium-ion battery (LIB) sector. Leveraging a unique dataset of corporate LIB patents, this study examines the co-evolution of spatial agglomeration and inter-city collaborative networks within China's LIB industry. Integrating spatial statistics, social network analysis, and geographic detectors across four sub-periods, we systematically track this reshaping process and identify its driving forces. Our findings reveal a dual trajectory of restructuring. Spatially, LIB R&D enterprises exhibit persistent east-west disparities, with innovation hotspots concentrated in coastal urban clusters, the Yangtze River Delta, Pearl River Delta, and Beijing-Tianjin-Hebei region. However, standard deviation ellipse analysis indicates a significant northward shift in the distribution center toward Ji'an, Jiangxi Province, suggesting gradual restructuring toward inland regions. In the network dimension, inter-city collaboration has expanded from 2 to 157 cities, yet overall connectivity remains low and fragmented. Notably, high-level connections concentrate among core cities unconstrained by geographical distance, indicating network structure is reshaped by node hierarchy rather than spatial proximity. Further analysis reveals that regional economic openness, industrial agglomeration, technological innovation capabilities, and policy support collectively shape enterprise distribution and network positions. By integrating spatial and network perspectives, these findings advance understanding of how strategic industries reshape regional innovation landscapes and provide evidence-based implications for fostering a more balanced and connected technological innovation landscape in China.

Keywords: Lithium-ion battery R&D firms; Battery industry; Energy industry; Spatial evolution; Collaborative innovation networks; Influencing factors

1 Introduction

The advancement of China's lithium-ion battery (LIB) industry plays a crucial role in achieving the country's peak carbon and carbon neutrality goals. In modern energy systems, energy storage systems represent a core sub-sector of the new energy industry chain, featuring independent technological pathways, business models, and innovation ecosystems. The effective use of clean energy depends on efficient energy storage systems. LIBs have been identified as the preferred energy storage solution for electric vehicles and renewable energy power generation systems (e.g., wind and solar energy) because of their high energy density, long cycle life, and relatively low cost. LIB energy storage not only enhances the supply stability of renewable energy but also creates favorable conditions for the energy system transition, thereby significantly advancing the achievement of United Nations Sustainable Development Goal 7 (SDG7: Ensure access to affordable, reliable, sustainable, and modern energy for all). Meanwhile, the aggressive development of the new energy vehicle industry has resulted in a dramatic growth in the global demand for LIB [1]. China's LIB production capacity accounted for more than 70% of the world's total in 2020, ranking first globally, according to Standard & Poor's (S&P) Global Market Intelligence. However, production dominance does not equate to technological leadership. China is not competitive in the core technology of LIB compared with developed countries such as the United States and Japan, which were early actors in this field [2]. Research and development (R&D) firms, as primary actors in regional innovation, promote the high-quality development of regional industries and enhance regional competitiveness [3, 4]. Therefore, clarifying how such enterprises achieve spatial organization and collaborative linkages is crucial for analyzing the innovation landscape of this strategic industry. However, the evolution of China's LIB R&D landscape and the forces that have reshaped it over time remain underexplored.

The spatial distribution of firms is an important topic in economic geography. Traditional location theories, such as agricultural location theory and central place theory, as well as modern location theories, such as new economic geography theory and heterogeneous firm trade theory, have laid a robust theoretical foundation for investigating the location selection of firms [5–8]. Empirical studies have also been conducted on the spatial distribution of firms using spatial statistical techniques, such as location entropy and standard deviation ellipse, based on data from logistics, creative, and fashion firms, among others [9–12]. As the field has progressed, more attention has been paid to factors affecting firm location selection. In this regard, the effects of firm attributes, e.g., size and innovation capabilities, as well as the external environment, e.g., regional market size, agglomeration economy, resource endowment, infrastructure, and policies, have been investigated using geographical weighted regression and Geographical Detector (GeoDetector), among other methods [13–20]. For example, Wang et al. [21] investigated the headquarters and subsidiaries of leading agricultural firms in Heilongjiang Province, China, using Moran's I and GeoDetector. They found a noticeable urban agglomeration of firm headquarters, and it was also determined that the location of the subsidiaries was significantly affected by farm and raw material sites.

Zhao and Miao [22] analyzed the point-of-interest (POI) data of logistics firms in Zhengzhou, China, from 2012 to 2021 using the industrial concentration index and kernel density. They found that the logistics industry developed rapidly in recent years, with obvious agglomeration in central cities, significant autonomy in spatial distribution, and increasing influence of industrial parks and traffic structure. Li and Zhang [23] analyzed the POI data of the urban service industry and tobacco retailers in Guanshanhu District of Guiyang, Guizhou Province, China, using the standard deviation ellipse approach. Their results revealed that this industry's spatial distribution mirrored that of service facilities, but it still had substantial room for development through the lens of population distribution. Wu et al. [24] investigated the spatial location of the urban cultural creative industry using a quantitative model and found that this industry exhibited a spatial pattern of spreading from the main urban area of Hangzhou, China, as the core and was evidently distributed around the provincial, municipal, and district government land. Beyond describing spatial patterns, researchers have increasingly focused on the determinants of firm location. By analyzing pollution-intensive firms' dynamic data, Song and Feng [25] found a noticeable spatial transfer of pollution-intensive industries in the Yangtze River urban agglomeration, with the central city as the core of the spatial distribution. Although these studies have greatly advanced our understanding of the spatial distribution of firms, they share a common limitation: an almost exclusive focus on static spatial structures. This static perspective is particularly inadequate for understanding strategic emerging industries such as the LIB industry. In emerging industries, innovation increasingly depends on the interaction between local agglomeration and cross-regional collaborative networks, and the dynamic mechanisms of the LIB industry remain underexplored in existing research.

The innovation network that emerges through inter-firm cooperation is an essential component of regional innovation [26]. Based on data covering collaborative innovation in patents, papers, and research projects, researchers have investigated the formation and evolution of innovation networks using social network analysis (SNA) from the perspective of global structural properties, such as average clustering coefficient and average path length, as well as in terms of node characteristics such as centrality and structural holes [27–31]. As the research field has developed, more academic efforts have examined the impact mechanisms of innovation networks, mainly in terms of endogenous effects, node attributes, and multidimensional proximity within network structures [32–34]. For example, in an SNA of patent data on high-tech industries, Song and Kim [35] identified stable innovation cooperation between cities in

the Yangtze River Delta with pronounced spatial spillover effects. Cai et al. [36] revealed noticeable “core–edge” characteristics and increasing density of the inter-provincial network for collaborative innovation in China based on patent data from 2008 to 2018, and further found that economic and R&D levels had a significant effect on the network. Based on data on Chinese cities from 2005 to 2017, Li et al. [37] demonstrated that cities classed as innovative had a sustained positive effect on the urban network for green innovation; in addition, green innovation was demonstrated to have a greater effect in the central and western regions of China. By examining biotechnology innovation papers and patents from 2000 to 2012, Li et al. [38] revealed that the scientific knowledge network was more complex than the technical knowledge network in terms of average degree and size. However, both of them were scale-free, and the drawbacks of cities in innovation can be better understood by comparing the two networks.

Therefore, the key theoretical gap lies in the analytical separation between spatial agglomeration and collaborative networks. Existing studies tend to examine these two dimensions in isolation, ignoring the fact that firm agglomeration and inter-firm networks are interdependent components of an evolving innovation system [39–42]. Spatial concentration facilitates knowledge spillovers and collaboration opportunities, while network connections can, in turn, reinforce or reshape spatial patterns. This co-evolutionary dynamic is particularly salient in strategic emerging industries such as the LIB industry, where technological breakthroughs often emerge from the interaction between local agglomeration and cross-regional collaboration. However, few studies have adopted an integrated framework to systematically track how spatial and network dimensions co-evolve over time and jointly shape the innovation landscape of strategic industries.

To address this gap, this study adopts an integrated analytical framework grounded in evolutionary economic geography and complex network theory. Using Chinese LIB patent data from the past two decades, we conceptualize the LIB R&D sector as a dynamically evolving technological innovation landscape, wherein spatial agglomeration and cross-city collaborative networks co-evolve. Specifically, we investigate: (1) the evolutionary trajectory of the spatial distribution of LIB R&D enterprises; (2) the development and restructuring of cross-city collaborative innovation networks; and (3) the driving factors behind the co-evolution of spatial patterns and network structures, and how they jointly reshape the innovation landscape. This study makes three original contributions. First, by integrating spatial and network perspectives, it moves beyond static, compartmentalized analyses to reveal the dynamic mechanisms through which agglomeration and collaboration mutually shape and co-evolve. Second, it conceptualizes cities as nodes in a dual-layered landscape, simultaneously embedded in geographic space and relational networks, and uncovers the interaction mechanisms between these layers. Third, using the geographical detector, it quantifies the relative importance of economic, innovation, and policy factors, providing empirical evidence for policy interventions aimed at fostering a more balanced, connected, and resilient LIB innovation landscape in China.

In summary, this study deepens the understanding of how the regional innovation landscape reshapes strategic emerging industries and provides actionable policy references for policymakers and industrial stakeholders.

2 Data and Methodology

2.1 Data

The data on Chinese LIB patents for the last 20 years were retrieved from the Incopat global patent database. Searching Incopat’s global patent database for Chinese LIB patents over the past two decades: IPC = H01M10/052, specifically retrieving patent families classified under H01M10/052. The selected timeframe spans 2004 to 2023, yielding 66,903 relevant patents after filtering and cleaning. Firms identified in the patents were then investigated using the Qixin Huiyan data platform (<https://b.qixin.com/>) to obtain basic information, such as their geographic coordinates and type. To accurately describe the evolution of the spatial distribution of LIB R&D firms in China and avoid dramatic changes caused by special circumstances in particular years, the study period was divided into four equal-length five-year subperiods based on application year: 2004–2008, 2009–2013, 2014–2018, and 2019–2023.

2.2 Methodology

2.2.1 Spatial statistical analysis

Spatial statistical analyses were conducted in ArcGIS to characterize the spatial distribution and clustering patterns of Chinese LIB R&D firms, including hot spot analysis (Getis–Ord G_i^*), spatial autocorrelation analysis (Moran’s I), and standard deviation ellipse analysis, following standard formulations [43, 44]. The hot spot analysis computes the Getis–Ord G_i^* statistic for each spatial unit and identifies locations where high or low values are significantly clustered, based on the corresponding z -scores and p -values.

2.2.2 Social network analysis

SNA was employed to examine the structural characteristics of the collaborative innovation network among LIB R&D firms in China. The network was constructed based on inter-firm collaborative R&D relationships derived from patent cooperation data. In this network, nodes represent LIB R&D firms, and edges represent collaborative innovation relationships between firms. The network is treated as an undirected and unweighted network, as collaborative

innovation implies mutual interaction between firms. To characterize the structural properties of the network, several standard network metrics were calculated using Gephi, including network density, average path length, degree centrality, and betweenness centrality. Network density measures the overall level of connectivity within the network and is defined as the ratio of the number of actual links to the maximum possible number of links [45–47]:

$$\text{Density} = \frac{2E}{N(N-1)} \quad (1)$$

where, N denotes the number of nodes in the network, and E represents the number of observed links. A higher density indicates a more interconnected collaborative network.

The average path length reflects the efficiency of information transmission within the network and is defined as the average number of steps along the shortest paths between all pairs of nodes:

$$\text{Average Path Length} = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij} \quad (2)$$

where, d_{ij} is the shortest path distance between nodes i and j . A smaller value of average path length indicates that information or knowledge can spread more efficiently across the network.

Degree centrality measures the number of direct connections a node has with other nodes, reflecting the level of participation of a firm in collaborative innovation activities. Betweenness centrality captures the extent to which a node lies on the shortest paths between other nodes, indicating the potential of a firm to act as an intermediary or broker in knowledge diffusion within the network.

These indicators jointly describe the connectivity, efficiency, and positional importance of firms within the collaborative innovation network [48–50].

2.2.3 Geographical detector

The effects of regional economy, innovation, and policies on the number of LIB R&D firms in individual cities and the centrality of the inter-firm collaborative innovation network were analyzed using GeoDetector. Specifically, the explanatory power of each factor was quantified using the q -statistic (factor detector):

$$q = 1 - \frac{\sum_{h=1}^L N_h \sigma_h^2}{N \sigma^2} \quad (3)$$

where, $h = 1, \dots, L$ denotes the strata of factor X , N_h and N are the sample sizes of stratum h and the whole study area, and σ_h^2 and σ^2 are the variances of Y , within stratum h and in the whole study area, respectively. A larger q indicates stronger explanatory power. This formulation follows the GeoDetector framework [50, 51].

3 Spatial Pattern of Lithium-Ion Battery Research and Development Firms in China

3.1 Overall Spatial Distribution of Lithium-Ion Battery Research and Development Firms

As shown in Figure 1, the number of LIB R&D firms in China exhibited an overall spatial pattern of decreasing from east to west. Specifically, more firms were located in the eastern coastal region, fewer in the central region, and even fewer in the western and northeastern regions. At the level of urban agglomerations, LIB R&D firms were concentrated in the Yangtze River Delta, Pearl River Delta, and Beijing–Tianjin–Hebei urban agglomerations. For LIB R&D activities, such locational advantages are particularly crucial, as battery innovation typically requires intensive capital investment, advanced manufacturing infrastructure, highly skilled engineering talent, and close interaction between upstream material suppliers and downstream application sectors such as electric vehicles and energy storage systems.

These patterns indicate that a favorable geographical location and high economic development and technological levels are critical factors in the location selection of Chinese LIB R&D firms. A favorable geographical location includes not only the absolute location (latitude and longitude) but also the relative location (relationship with surrounding areas). A favorable geographical location usually refers to factors such as a city's terrain, climate, and its transportation connections with other areas. For example, cities located in the plains and coastal areas of China generally have better resources and more developed industries due to the unique geographical advantages of these areas. The average nearest neighbor index was calculated to be 0.1583 and was significant at the 1% level, demonstrating the clustered distribution of LIB R&D firms in China.

Remarkable differences were found in the number of LIB R&D firms at different geographic levels. At the provincial level, Guangdong and Jiangsu had the largest number of LIB R&D firms (2,503 and 1,690, respectively), followed by Zhejiang and Anhui (788 and 517, respectively), as shown in Figure 2a. Nine provinces, including Shandong, Shanghai, Hunan, and Beijing, had more than 200 firms, whereas 14 provinces, including Guangxi,

Liaoning, Guizhou, and Inner Mongolia, had fewer than 100 firms. This indicates that there is a high degree of clustering at the provincial level, and a small number of provinces are home to core agglomerations.

At the municipal level, 21 cities, including Shenzhen, Suzhou, Dongguan, Shanghai, Beijing, Huizhou, and Changzhou, had more than 100 LIB R&D firms. Shenzhen had the highest number of firms (1,177), followed by Suzhou and Dongguan (more than 500), as shown in Figure 2b. A further sixteen cities, including Zhuhai, Foshan, Qingdao, Xi'an, and Xuzhou, had more than 50 firms. In total, these 37 cities accounted for 70.30% of China's LIB R&D firms. Meanwhile, 107 cities had five or fewer firms, accounting for 37.41% of the total number of Chinese cities with LIB R&D firms. These findings again indicate a high level of clustering of LIB firms in China.

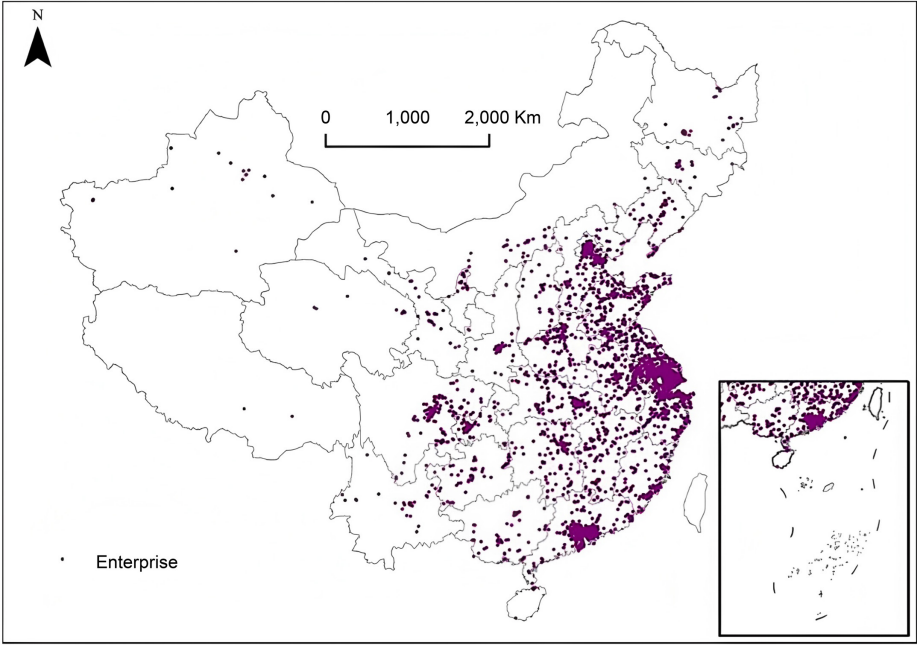


Figure 1. Spatial distribution of lithium-ion battery research and development (LIB R&D) firms in China

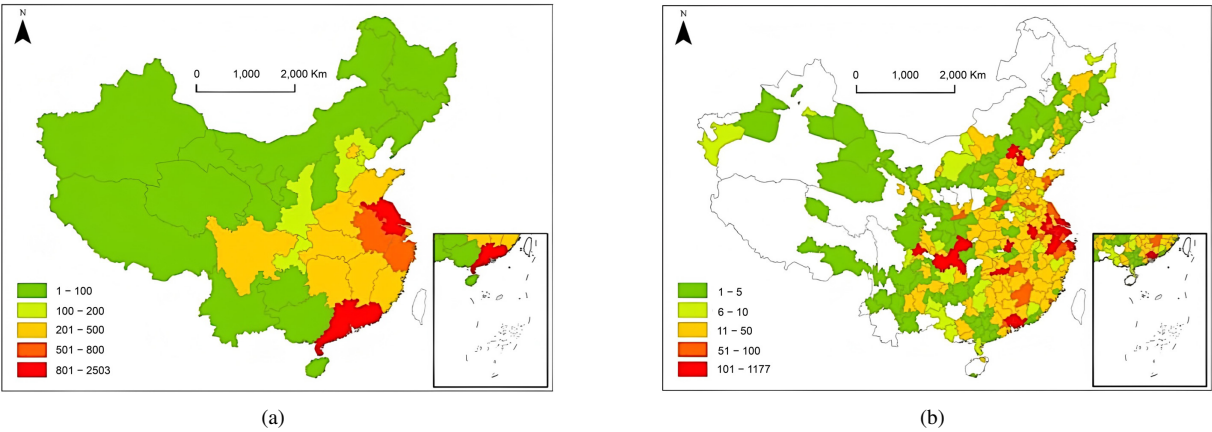


Figure 2. Spatial distribution of Chinese lithium-ion battery research and development (LIB R&D) firms at different levels: (a) Provincial level; (b) Municipal level

3.2 Spatial Agglomeration of Lithium-Ion Battery Research and Development Firms

The standard deviation ellipses of LIB R&D firms were calculated for each sub-period using ArcGIS. The results are presented in Table 1 and Figure 3. It can be seen that the standard deviation changed markedly over the last 20 years. The location areas have a wide geographic dispersion, presenting a northeast–southwest pattern of spatial distribution. The major axis standard deviation decreased, but the minor one increased, indicating the sustained spread of LIB R&D firms in the northeast–southwest and northwest–southeast directions. This distribution pattern

indicates that China's LIB industry is forming an innovation corridor with distinct regional characteristics, linking coastal manufacturing hubs or R&D centers with inland resource-rich areas. In terms of the distribution center, the ellipse center shifted northward from Ji'an to Fuzhou in Jiangxi Province, and finally moved further northward to Jiujiang in Jiangxi Province.

Table 1. Standard deviation ellipse parameters of lithium-ion battery research and development (LIB R&D) firms in China

Years	Center Longitude	Center Latitude	Major Axis Standard Deviation (km)	Minor Axis Standard Deviation (km)	Rotation Angle (°)	Center Location
2004–2008	115.3912	26.9682	99.7805	30.4725	6.3739	Ji'an Jiangxi Province
2009–2013	115.9871	27.7650	92.4630	40.8333	8.6332	Fuzhou, Jiangxi Province
2014–2018	115.7342	29.6608	86.9170	56.2576	6.8297	Jiujiang, Jiangxi Province
2019–2023	116.0751	29.4878	84.7017	52.1261	9.4059	Jiujiang, Jiangxi Province

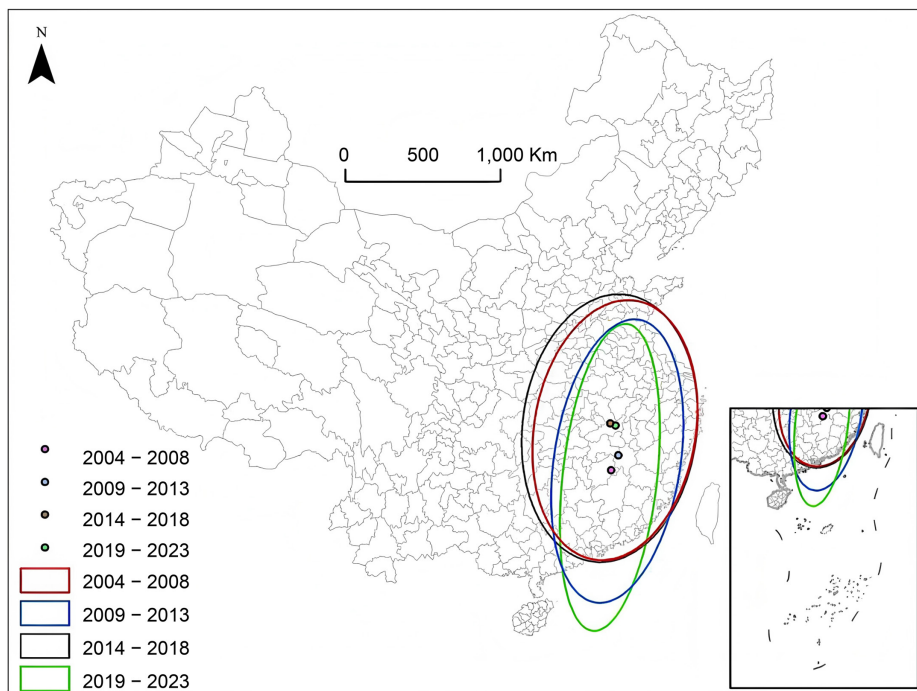


Figure 3. Standard deviation ellipses of lithium-ion battery research and development (LIB R&D) firms in China

In general, there was a considerable shift in the location distribution of LIB R&D firms in China. This may be due to the regional differences in the endowment of lithium resources across China. The eastern region builds major production bases by leveraging its industrial chain and technological advantages, whereas the central and western regions rely on huge reserves of lithium resources to develop primary industries. Acquiring lithium resources not only reduces input costs for upstream battery material production but also promotes the coordinated development of resource extraction, material processing, and application R&D activities, thereby strengthening the integration of regional battery supply chains. Industrial upgrading, increasing market demand, and the guidance of national policies have promoted the formation of a coordinated development pattern with the eastern region of China as the leader and the central and western regions being driven.

Spatial autocorrelation analysis using ArcGIS (Table 2) revealed that Moran's I was always greater than zero and significant at the 1% level, which indicates significant spatial autocorrelation of the LIB R&D firms. In other words, cities with a large number of LIB R&D firms had neighboring cities with a relatively large number of LIB R&D firms.

Moreover, the increasing value of Moran's I suggests that the spatial autocorrelation of the LIB R&D firms increased.

Table 2. Spatial autocorrelation of lithium-ion battery research and development (LIB R&D) firms in China

Years	Moran's I	<i>p</i> -Value	<i>z</i> -Value
2004–2008	0.1161	0.000	11.0658
2009–2013	0.1568	0.000	13.5990
2014–2018	0.2052	0.000	17.1051
2019–2023	0.2157	0.000	17.9825

Furthermore, local hotspot detection was performed using ArcGIS to reveal the hotspot changes in the spatial distribution of LIB R&D firms from 2004 to 2023 (Figure 4). This LIB innovation activity thereby highlights the geographic concentration of battery-related R&D capabilities. Shenzhen was the only city that remained a hotspot area throughout the study period. It was the only hotspot until 2009–2013, when Dongguan, Suzhou, Shanghai, and Beijing were identified as newly emerged hot areas. Shenzhen, as a longstanding hub for LIB innovation, demonstrates its sustained leadership in this field within China. This advantage stems from its robust electronics manufacturing base, dense R&D network, and close integration with downstream energy storage and electric vehicle industries. The sub-hot areas were concentrated in Guangzhou, Huizhou, and Dongguan in the Pearl River Delta, Hangzhou, Suzhou, and Shanghai in the Yangtze River Delta, and Beijing in the Beijing–Tianjin–Hebei region. Specifically, nine cities, including Guangzhou, Huizhou, Dongguan, Hangzhou, and Suzhou, were classified as sub-hot areas in 2004–2008; nine cities, including Guangzhou, Huizhou, Hangzhou, Huzhou, and Ningbo, were so classified in 2009–2013; Dongguan, Suzhou, Shanghai and Beijing were thus classified in 2014–2018; and 10 cities, including Guangzhou, Huizhou, Hangzhou, Shanghai, and Beijing, were classified as sub-hot in 2019–2023. These changes should be understood in the context of Shenzhen further widening the gap with other cities, such as Suzhou, Shanghai, and Beijing, rather than Beijing being downgraded.

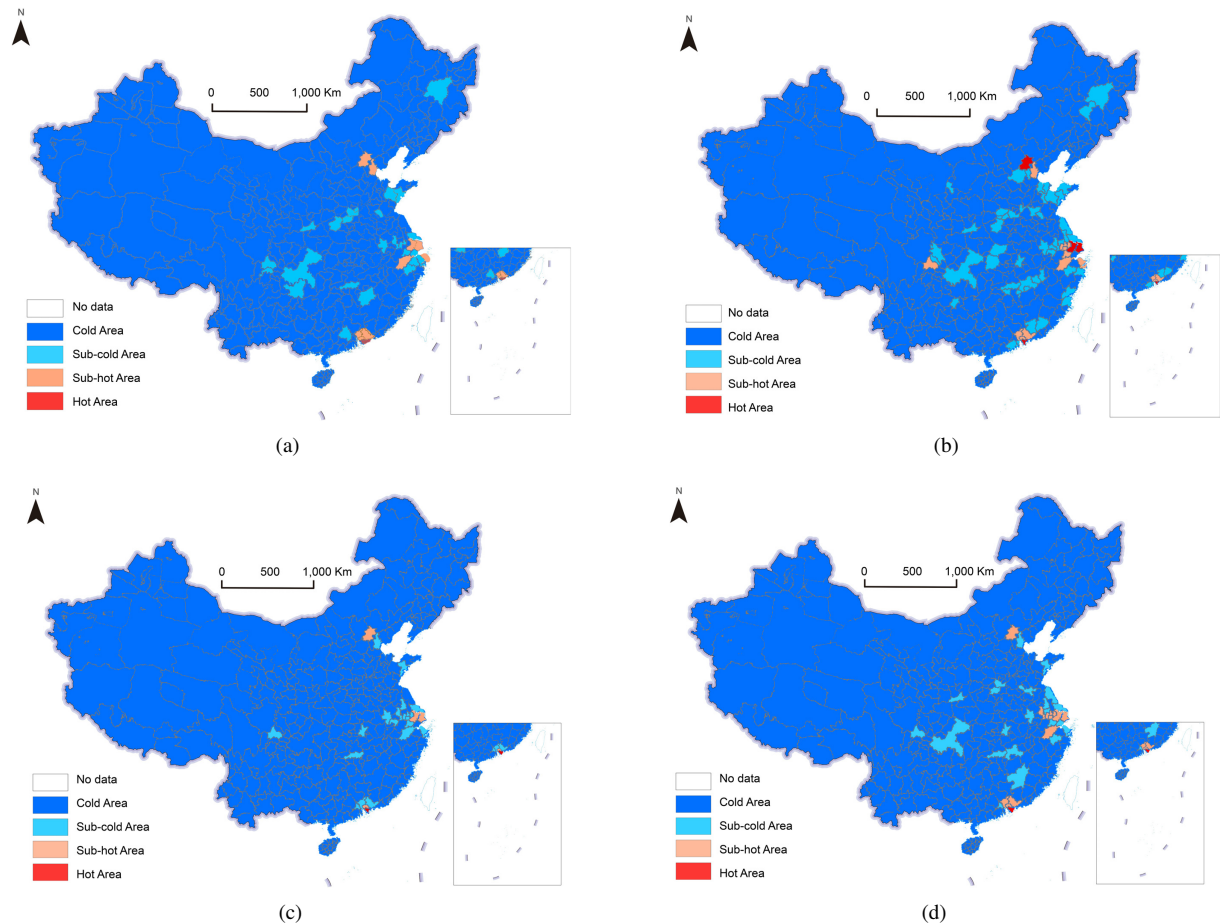


Figure 4. Spatial distribution of hot and cold areas of lithium-ion battery research and development (LIB R&D) firms in China: (a) 2004–2008; (b) 2009–2013; (c) 2014–2018; (d) 2019–2023

In general, the distribution of the hot and cold areas remained relatively stable. The hot areas were mostly distributed in the Pearl River Delta and Yangtze River Delta urban agglomerations along the eastern coast, whereas the cold areas were mainly distributed in the western, northeastern, and central regions, which were weakly driven by the eastern coastal region. The spatial distribution of LIB R&D firms in China is closely related to the regional economy, innovation, and local government support, which is similar to previous findings [52, 53]. From a battery industry perspective, these findings suggest that coordinated policy interventions and cross-regional collaboration mechanisms are essential for fostering a more inclusive and resilient LIB innovation ecosystem.

3.3 Collaborative Innovation Network Characteristics of Lithium-Ion Battery Research and Development Firms

A total of 6,584 LIB patents with ≥ 2 firms as applicants were extracted during the study period. These jointly applied patents represent formalized technological collaboration activities within China's LIB industry and provide a reliable proxy for inter-firm R&D cooperation intensity. For example, a patent jointly applied for by firms A, B, and C generates three pairwise collaboration links (A–B, A–C, and B–C). As a pragmatic weighting scheme to differentiate primary collaboration ties from secondary co-application links, the link between the first and second applicants was assigned a weight of 2, whereas links involving other co-applicants were assigned a weight of 1. The collaborative innovation network of LIB R&D firms was established by retrieving cities where the applicants were registered from the Qixin Huiyan platform.

The inter-firm collaborative innovation network was analyzed using Gephi software (Table 3). Results showed that the number and connections of LIB R&D firms increased with increasing market demand. This trend aligns with the rapid expansion of the electric vehicle and energy storage markets, spurring deeper technological collaboration among battery-related enterprises. The number of participating cities increased from 2 to 157, and the number of connection edges also increased from 1 to 515 over the analyzed timeframe. The average degree and average weighted degree of the network increased from 0.5 and 1 in 2004–2008 to 3.25 and 66.28 in 2019–2023, respectively.

Table 3. Collaborative innovation network attributes of lithium-ion battery research and development (LIB R&D) firms

Network Attributes	2004–2008	2009–2013	2014–2018	2019–2023
Node	2	25	15	157
Number of connections	1	29	159	515
Average degree	0.50	1.16	1.83	3.28
Average weighted degree	1	15.92	26.24	66.28
Network diameter	1	3	8	7
Network density	0.500	0.048	0.021	0.021
Average clustering coefficient	0	0.075	0.134	0.260
Average path length	1	1.802	3.459	3.037

These findings indicate that the network structure became increasingly complex, while the network size and correlation increased significantly. However, the network density decreased over time, which suggests weak inter-city connections. The average clustering coefficient and average path length increased significantly, indicating decreasing connectivity and marked network dispersion.

The network connection weights and centrality in 2019–2023 were classified into five and three levels, respectively, according to natural breaks. Natural breaks offer a data classification method that divides the data into several regions by calculating the variable value of each breakpoint. Thus, their use allowed minimizing the intra-regional differences in data values and maximizing inter-regional ones. Classifications were also determined for 2004–2008, 2009–2013, and 2014–2018 using the boundaries set in 2019–2023 (Figure 5). Using a consistent classification benchmark across periods allows for a longitudinal comparison of structural changes in the battery innovation network, rather than reflecting short-term fluctuations. In terms of node centrality, only two cities, Shanghai and Hangzhou, were at the third level in 2004–2008. This indicates that China's LIB R&D collaboration network initially relied on a very limited number of metropolitan innovation hubs. Similarly, only Dongguan and Ningde were at the second level in 2009–2013. In 2014–2018, 11 additional cities, including Shanghai, Wuxi, Shenzhen, and Beijing, were upgraded to the second level.

In 2019–2023, Shanghai, Wuxi, Shenzhen, Beijing, Dongguan, and Nanjing were upgraded from the second level to the first level. Two additional cities, Foshan and Changsha, were also classified in the first level. Meanwhile, 25 additional cities, including Hangzhou and Mianyang, were classified as reaching the second level.

Overall, the findings indicate significant improvement in city levels over time, accompanied by an increasing number of high-level cities and their spread to the central and western regions. There were more high-level cities in

the eastern region and fewer in the central and western regions, exhibiting a spatial pattern of strength in the east and weakness in the west.

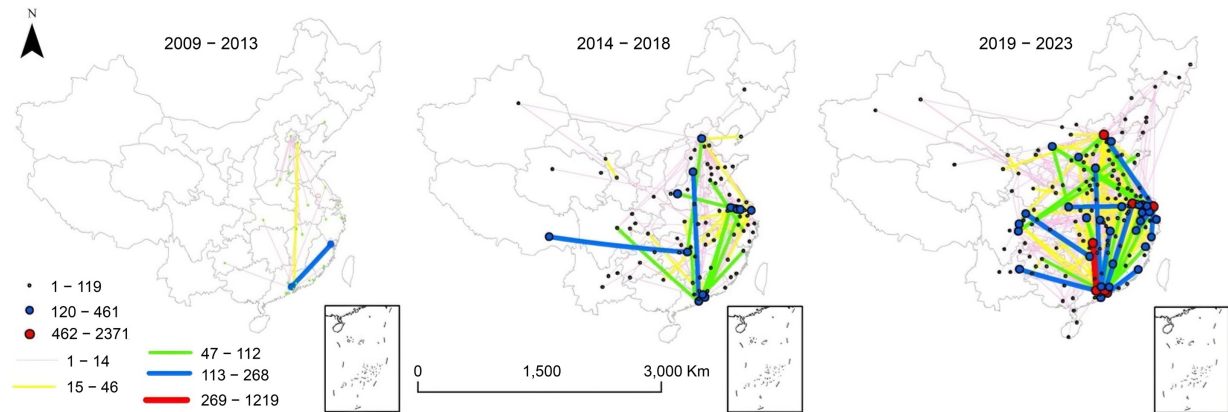


Figure 5. Collaborative innovation network of lithium-ion battery research and development (LIB R&D) firms in China

Note: In 2004–2008, there was only one pair of connected cities, Shanghai–Hangzhou, with a connection strength of 2. Both cities had a centrality of 2. Therefore, this sub-period is not illustrated here.

In terms of network connections, in 2004–2008, only one pair of connected cities, Shanghai–Hangzhou, was identified, with a connection strength of 2. In 2009–2013, only Ningde–Dongguan had attained the second level; there were no third-level connections, and Tianjin–Shenzhen was at the fourth level. In 2014–2018, Linzhi–Xiangtan and Handan–Zhuhai had reached the second level, and 12 pairs of cities, including Beijing–Xiangtan, Foshan–Changsha, and Huizhou–Shenzhen, attained the third level.

In 2019–2023, Wuxi–Shanghai and Foshan–Changsha had the strongest connection and were at the first level. Meanwhile, 12 pairs of cities, including Dongguan–Nanjing, Shenzhen–Xiamen, and Beijing–Shanghai, reached the second level; and 29 pairs of cities, including Hangzhou–Huzhou and Ningbo–Hangzhou, were at the third level. On the whole, there were noticeable spatial differences in connection strength, and the levels of connection significantly improved over time. The city connections were sparse in the central and western regions but dense in the eastern coastal region. Interestingly, network connections were not noticeably constrained by distance. High-level connections primarily occurred between core node cities rather than between a core node and a closer but less developed city. The collaborative innovation network of LIB R&D firms in China was profoundly affected by local economic level and innovation capabilities, which resonates with previous findings [54].

4 Factors Affecting the Spatial Distribution of Lithium-Ion Battery Research and Development Firms in China

4.1 Variable Selection

Numerous factors affect the spatial distribution and evolution of LIB R&D firms. By combining location theory and geographical perspectives, the factors affecting the spatial distribution of LIB R&D firms are categorized into regional economy, innovation, and policies based on previous studies [55–59]. Given data availability, in terms of the regional economy, the analysis included (1) total retail sales of consumer goods representing household consumption, (2) total imports and exports representing openness, (3) regional gross domestic product (GDP) per capita representing economic level, and (4) the number of industrial enterprises above a designated size representing industrial clustering. In terms of regional innovation, the number of colleges and universities representing university resources and the number of patents representing technological innovation were selected. In terms of regional policies, technology expenditure as a percentage of GDP was included to represent government support, year-end outstanding Renminbi (RMB) loans of financial institutions were used to represent financial system support, the number of development zones was used to represent development zone policy, and the city administrative level was taken to represent political resources.

Regarding the administrative classification of cities, a value of 1 is assigned to general cities, 2 to sub-provincial cities, 3 to provincial capitals, 4 to Shanghai, Chongqing, and Tianjin, and 5 to Beijing. With regard to colleges and universities, a value of 2 is assigned to universities included in the “double world-class project,” and 1 to general colleges and universities. Regarding development zones, national development zones are assigned a value of 2, and provincial ones are assigned a value of 1.

The number of LIB R&D firms in individual cities and the centrality of the inter-firm innovation network in China in 2019–2023 were used as the dependent variables. They were classified into five levels by natural breaks

using ArcGIS before GeoDetector was employed. Data on the influencing factors came from China City Statistical Yearbooks and the statistical yearbooks of relevant provinces.

4.2 Results and Driving Factors

GeoDetector analysis (Table 4) revealed that regional economy, innovation, and policy factors have significant effects on the spatial distribution of LIB R&D firms across cities and also shape the weighted network centrality of inter-firm collaboration in China. For the number of LIB R&D firms, the strongest determinants are openness ($q = 0.5400$, $p < 0.001$), technological innovation measured by patents ($q = 0.5047$, $p < 0.001$), and industrial clustering ($q = 0.4515$, $p < 0.001$). For weighted network centrality, the top explanatory factors are technological innovation ($q = 0.4730$, $p < 0.001$), openness ($q = 0.4689$, $p < 0.001$), and industrial clustering ($q = 0.4059$, $p < 0.001$). These results highlight the high degree of global integration and technological complexity of the LIB industry, which relies heavily on international knowledge exchange, supply-chain coordination, and concentrated industrial ecosystems. Overall, market-driven and innovation-oriented forces appear to outweigh purely administrative advantages in shaping both firm location and collaboration prominence within the LIB R&D network.

Table 4. Analysis of influencing factors

Dimension	Variable	Measure	Number of LIB R&D Firms		Weighted Network Centrality	
			<i>q</i> -Value	<i>p</i> -Value	<i>q</i> -Value	<i>p</i> -Value
Regional policies	Household consumption	Total retail sales of consumer goods	0.3668	0.0000	0.3052	0.0000
	Openness	Total imports and exports	0.5400	0.0000	0.4689	0.0000
	Economic level	Regional GDP per capita	0.3454	0.0000	0.3165	0.0000
	Industrial clustering	Number of industrial enterprises above designated size	0.4515	0.0000	0.4059	0.0000
Regional innovation	University resources	Number of colleges and universities	0.1963	0.0000	0.2004	0.0000
	Technological innovation	Number of patents	0.5047	0.0000	0.4730	0.0000
Regional policies	Government support	Technology expenditure as a percentage of GDP	0.3022	0.0000	0.2331	0.0000
	Financial system support	Year-end outstanding RMB loans of financial institutions	0.3899	0.0000	0.3930	0.0000
	Development zone policy	Number of development zones	0.1832	0.0000	0.1750	0.0023
	Political resources	City administrative level	0.1826	0.3176	0.2088	0.1655

Note: LIB R&D = lithium-ion battery research and development; GDP = gross domestic product; Renminbi = RMB.

In terms of the regional economy dimension, economic level, household consumption, openness, and industrial clustering all exhibit statistically significant explanatory power for both dependent variables (Table 4; $p < 0.001$). This suggests that stronger local economic fundamentals and more open, agglomerated industrial environments are associated with larger concentrations of LIB R&D firms and more central positions in the collaboration network. Economic level provides an important basis for firms' regional R&D and innovation activities, while higher household consumption, openness, and industrial clustering may help reduce innovation frictions and coordination costs by improving market demand, external linkages, and localized supply-chain support. In particular, the positive externalities and resilience generated by industrial clustering can facilitate the development of LIB R&D firms and strengthen their collaborative connections.

Within the regional innovation dimension, patent-based technological innovation exhibits strong explanatory power for both firm distribution and weighted network centrality ($q = 0.5047$ and 0.4730 , respectively; $p < 0.001$). University resources, measured by the number of colleges and universities, also contribute, although with lower q -values ($q = 0.1963$ for firms and 0.2004 for centrality; $p < 0.001$). These results suggest that cities with stronger innovation capacity are more likely to attract and sustain LIB R&D activity and occupy more central positions in collaboration networks. Colleges and universities can facilitate industry–university–research cooperation by providing skilled talent and knowledge spillovers, supporting key technological breakthroughs in materials science, electrochemistry, and energy storage [60].

Within the policy dimension, government support, financial system support, and development-zone policy are statistically significant and show moderate explanatory power for both dependent variables (Table 4; $p < 0.01$). Local

fiscal and policy support, together with access to financial services, can alleviate constraints on R&D and innovation activities, while development zones provide institutional and physical platforms that facilitate firm innovation and growth. By contrast, political resources proxied by city administrative level show no significant explanatory effect for either dependent variable (Table 4; $p = 0.3176$ for the number of firms; $p = 0.1655$ for weighted centrality), indicating that administrative hierarchy plays a limited role in this context.

This pattern may be linked to China's broader transition toward market-oriented allocation mechanisms since the reform and opening up. As many non-core cities have rapidly improved their economic foundations and innovation capacity, the traditional advantages associated with higher administrative status appear to have weakened, making openness, industrial agglomeration, and innovation capacity more decisive for attracting LIB R&D activity and shaping collaboration networks. In this context, policy tools such as development zones may matter primarily through the resources and platforms they provide, rather than through administrative hierarchy itself.

5 Conclusions and Policy Implications

5.1 Conclusions

The analysis was conducted using spatial statistical techniques, SNA, and GeoDetector based on data of Chinese LIB patents with firm applicants from 2004 to 2023. The following findings are established. First, the number of LIB R&D firms generally presented a spatial pattern of decreasing from the east to the west in China. They were concentrated in the Yangtze River Delta, Pearl River Delta, and Beijing–Tianjin–Hebei urban agglomerations. At the provincial level, Guangdong and Jiangsu had the largest number of LIB R&D firms. At the municipal level, 21 cities, including Shenzhen and Suzhou, had more than 100 LIB R&D firms.

Second, the standard deviation ellipse analysis revealed that the location areas of LIB R&D firms were dispersedly distributed in China, presenting a northeast–southwest pattern of spatial distribution. Overall, there was a substantial shift in the location distribution over the study timeframe, with a coordinated development pattern emerging with the eastern region as the leader and the central and western regions being driven. Spatial autocorrelation analysis demonstrated a significant and increasing spatial correlation between the LIB R&D firms. The distribution of hot and cold areas remained relatively stable. The hot areas were primarily distributed in the Pearl River Delta and Yangtze River Delta urban agglomerations along China's eastern coast.

Third, SNA revealed that the structure of the collaborative innovation network of China's LIB R&D firms became increasingly complex. The network size and correlation increased significantly, but with decreasing connectivity and noticeable dispersion. The city levels significantly improved over time, accompanied by an increasing number of high-level cities and their spread to the central and western regions, and presenting a spatial pattern of strength in the east and weakness in the west. Marked spatial differences were found in network connection strength, and the levels of connection significantly increased over time. The city connections were sparse in the central and western regions and dense in the eastern coastal region. In addition, network connections were not noticeably constrained by distance. High-level connections primarily occurred between core node cities.

Fourth, GeoDetector analysis suggested that the distribution of LIB R&D firms across cities and the inter-firm innovation network in China were significantly affected by regional economy, innovation, and policies.

5.2 Implications and Recommendations

First, it is advisable to establish a collaborative innovation network and create an environment that fosters industry–university–research cooperation for innovation. The analysis of influencing factors reveals that regional innovation is a crucial factor in the spatial distribution and connection of LIB R&D firms. Therefore, local governments should foster the development of universities included in the “double world-class project,” establish a platform for industry–university–research innovation cooperation, cultivate innovative talent, create a favorable atmosphere for innovation, and introduce university resources to R&D firms through the platform to boost innovation.

Second, the government should play a role in supporting LIB R&D firms for innovation. Specifically, the government should plan, make adequate financial investments in technology, enhance the financial system support, promote the implementation of national and provincial development zone policies, and boost the innovation activities of LIB R&D firms in the development zones, thereby enhancing governmental service efficiency and support.

Third, efforts should be made to strengthen the connections between LIB R&D firms in different cities. Given the weak connection between cities in the network of China's LIB R&D firms, it is necessary to create a cooperation model for LIB R&D innovation with high-level cities, such as Shanghai, Wuxi, Shenzhen, and Beijing, as the core to radiate to other cities, thereby boosting cooperation between cities.

5.3 Theoretical Contribution

First, LIB patents with firm applicants were used as the data source of urban network from the perspective of evolutionary economic geography, which complements research on cities with different functions in the industrial network and guides the development of the Chinese LIB industry. By constructing an inter-city collaborative

innovation network based specifically on LIB patents, this study moves beyond generic urban network analyses and provides an industry-specific empirical framework for examining the structural organization of battery innovation systems.

Second, spatial statistical techniques like the standard deviation ellipse were employed to delve into the spatiotemporal evolution of the number of LIB R&D firms in China, which helps understand the status quo of the Chinese LIB industry and extends the scientific aspect of provincial and municipal geographies. Rather than merely describing spatial distribution patterns, the integration of spatial statistical tools with network analysis reveals the directional shift, hierarchical restructuring, and diffusion trajectory of the LIB innovation landscape.

Third, firm innovation is indispensable to the development of the LIB industry. As a carrier of technology development, R&D firms play an essential role in the exchange of firm innovation elements across cities. In this context, this study expands the geographical research on urban networks dominated by the knowledge economy by investigating the innovation cooperation between LIB R&D firms. More importantly, this study reveals the unique mechanisms of hierarchical agglomeration and selective connectivity in inter-firm collaboration within China's LIB sector, thereby deepening the theoretical understanding of battery innovation systems.

5.4 Limitations and Directions for Further Research

First, the location selection of LIB R&D firms is a complex issue. However, this study only considered regional economies, innovation, and policies in the analysis of influencing factors. In fact, R&D innovation also faces challenges in terms of commercialization and other aspects. Therefore, local R&D innovation costs are also a major factor of concern for firms regarding location selection, which should be an important direction in future research.

Second, the LIB R&D firms analyzed in this study were limited to patent applicants. Future research can be expanded to cover LIB R&D firms extracted from academic papers and those identified by business scope, and other sources.

Author Contributions

Conceptualization, F.H. and J.H.H.; methodology, H.J.Y. and J.Y.C.; software, S.B.W. and L.P.Q.; validation, L.P.Q. and S.Z.; formal analysis, H.Y.Z. and S.B.Y.; investigation, X.P.W. and J.Y.C.; resources, X.P.W.; data curation, L.P.Q.; writing—original draft, F.H. and J.H.H.; writing—review and editing, S.Z., H.J.Y., and H.H.; visualization, S.B.W.; supervision, J.Y.C. and F.H.; project administration, H.H. and F.H.; funding acquisition, F.H. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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